CAO Yiyang

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EDUCATION BACKGROUND

National University of Singapore (NUS) **Aug 2025 - Dec 2026 (expected)**

- MSc in **Mechanical Engineering** | GPA: **4.57/5.0**
- **Thesis:** Gradient-Induced Softening at Diamond--Gyroid Transitions in Ti-6Al-4V TPMS Lattices: Deformation Mechanisms, Failure Modes, and Surrogate-Assisted Optimization
- Relevant Courses: *Additive and Non-Conventional Manufacturing Processes (A), Engineering Materials in Medicine (A), Corrosion of Materials, Data-Driven Engineering and Machine Learning*

NUS (Suzhou) Research Institute (NUSRI) **Sep 2024 - May 2025**

- Selective Exchange Program | GPA: **86.23/100**

Soochow University (SUDA) "Project 211" **Sep 2021 - Jun 2025**

- B.Eng. in **Intelligent Manufacturing Engineering** & **Computer Science and Technology**
- GPA: **3.6/4.0**
- Course Highlight: *Engineering Materials (90), Artificial Intelligence (98), Fundamentals of Mechanical Design (97), Engineering Mechanics (91), Advanced Mathematics (90)*

RESEARCH INTERESTS & PROFILE

Experience in machine learning, finite element analysis, and materials testing.
Selected projects in computational design and machine learning are available on my GitHub portfolio.
Interests: intelligent manufacturing, soft-material biofabrication, and acoustic levitation 3D printing.

RESEARCH EXPERIENCE

Research Assistant, School of Advanced Materials, Peking University Shenzhen Graduate School
Advised by Prof. Zhaolong Chen, Peking University **Aug 2026 - Present**

- Conducted MoS_2 growth trials on sapphire substrates, varying applied-voltage magnitude and polarity, heating schedules, and sulfurization duration
- Performed Raman characterization of as-grown samples; ongoing work focuses on screening growth conditions for well-defined triangular MoS_2 domains

Gradient-Induced Softening at Diamond-Gyroid Transitions in Ti-6Al-4V TPMS Lattices: Deformation Mechanisms, Failure Modes, and Surrogate-Assisted Optimization **Aug 2025 – Present**
Advised by Prof. SWEE Leong Sing, Highly Cited Researcher, NUS

- Conducted finite element simulations and compression testing of LPBF-fabricated Ti-6Al-4V graded TPMS lattices to investigate deformation and failure mechanisms for orthopedic implants
- Developed a surrogate-assisted optimization framework combining Gaussian Process Regression (GPR) and NSGA-II to tailor lattice stiffness toward the cortical bone modulus range
- Developed and open-sourced a TPMS design platform for automated generation of hybrid architectures and simulation-ready STL geometries (GitHub)
- Identified a topology transition softening effect (TTSE) in Diamond-Gyroid hybrid lattices that enables tunable stiffness and failure behavior for mitigating stress shielding in porous implants

Data-Driven Prediction of Relative Density in LPBF 316L Stainless Steel via Multi-Source Dataset Integration and Neural-Network Modeling **Oct 2024 - Apr 2025**
Advised by Prof. SWEE Leong Sing, Highly Cited Researcher, NUS

- Built a multi-source LPBF 316L dataset by integrating literature data with in-house fabrication and density measurements to improve process-parameter coverage
- Compared the predictive performance of multiple machine-learning models for estimating relative density from LPBF process parameters

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Design and Implementation of an Automated Parser for Multi-source Heterogeneous Log Data

Advised by Prof. CHAO Pingfu, School of Computer Science and Technology, SUDA **Jan 2024 - May 2024**

- Integrated an existing Drain parser with preprocessing and evaluation modules into a Python workflow for heterogeneous system and application logs.
- Implemented Bayesian optimization to tune Drain's similarity threshold and tree depth, comparing parsing accuracy and F1-score against a fixed-parameter baseline across 16 Loghub datasets.

Computational Tool Optimization for Materials Analysis **Oct 2023 - Jan 2024**

Advised by Prof. CHAO Pingfu & Dr. LIAO Fan, SUDA

- Enhanced Python-based TPV analysis software using SciPy and Matplotlib, implementing peak-height thresholds and automated segmentation to limit false splits from minor peaks, alongside FFT spectral analysis and graphical export.

COMPETITIONS & ACTIVITIES

Team Leader| Kaggle: Large Language Model Prompt Recovery Competition **May 2024**

- Applied a Mistral 7B inference pipeline adapted from a public baseline, combining few-shot predictions of rewriting instructions and stylistic attributes (Code)
- Earned a silver medal, ranking 95th out of 2,175 teams (top 4.4%) (Certificate)

Third Prize | 18th Siemens Cup China Intelligent Manufacturing Challenge (CIMC) **Aug 2024**

- Built a production line digital twin system using Siemens industrial software for simulation, process modeling, and optimization analysis

Second Prize | CCB Cup Jiangsu Undergraduate Innovation Competition **Jul 2024**

- Designed process optimization solutions for ultra-precision polishing in semiconductor fabrication

National Project, Undergraduate Innovation and Entrepreneurship Training Program **May 2024**

- Provided digital media solutions for Suzhou classical gardens in a national-level project collaborated with the School of Arts, Soochow University

Exchange Program | University of Cambridge **Aug 2023**

- Participated in the Leadership and Management, Creativity and Innovation program

Student Ambassador, NUS (Suzhou) Research Institute **Sep 2024 - May 2025**

- Promoted student engagement through outreach activities and event planning

HONORS & AWARDS

Outstanding Graduate (top 10%), Soochow University	Jun 2025
Excellent Campus Ambassador, NUS Suzhou Research Institute	May 2025
Comprehensive Scholarship (top 15%), Soochow University	Dec 2023
Second-Class Academic Excellence Scholarship (top 15%), Soochow University	Dec 2023
Second Prize, Kaixuan Mechanical and Electrical Award, Soochow University	May 2023

SKILLS

- **Materials Fabrication & Characterization:** LPBF/SLM metal additive manufacturing, Scanning Electron Microscopy (SEM), Universal Testing Machine (quasi-static compression / tensile)
- **Engineering Software:** CAD/CAM (SolidWorks), Numerical Simulation (ANSYS, Abaqus)
- **Programming/Algorithms:** Java, Python (PyTorch, TensorFlow), SQL, MATLAB
- **English:** IELTS 7.0